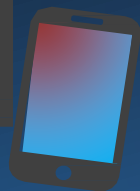




RPCA Cyber Club
Faculty of Forensic Science
Royal Police Cadet Academy

Cybersecurity

Chapter 8



OSINTs for Investigations

Asst.Prof.Pol.Lt.Col.Dr.Wongyos Keardsri (Bank)

Lecturer (Level 3) | Faculty of Forensic Science | Royal Police Cadet Academy | Thailand

Email: wongyos@gmail.com | Line ID: wongyos | Phone/WhatsApp: +6689-599-3490

Facebook: facebook.com/wongyos

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OSINT Overviews

1





Definition of OSINT

- **Open-Source Intelligence (OSINT)** refers to the process of **collecting, analyzing, and making decisions** of publicly available information (qualitative, quantitative) from **open sources** to produce actionable **intelligence**
- It is widely used in **cybersecurity, law enforcement, military operations, and competitive business analysis**



<https://www.techslang.com/definition/what-is-osint/>



Role of OSINT in Cyber Investigations

- In cyber investigations, **OSINT** plays a critical role in **identifying digital identities, tracing infrastructure, verifying online behavior, and building investigative leads**
- **OSINT** is a foundational **intelligence layer** that supports **cyber investigations** by:
 - Identifying **digital identities**
 - Mapping **online infrastructure**
 - Verifying **behavioral patterns**
 - Discovering **relationships** between **entities**
 - Supporting **attribution efforts (original sources)**



Sources of OSINT

■ Human-centric sources:

- Social media platforms
- Public records
- Online communities

■ Technical sources:

- Domain/IP information
- Metadata
- Digital archives

■ Media sources:

- News and blogs platforms
- Multimedia content

■ Specialized databases:

- Blockchain explorers
- Leak repositories
- Historical web archives



Public Records



Images/Videos



Websites



Social Media
Platforms



News Media



Libraries



OSINT Intelligence Cycle

- **Planning:** Define investigative objectives and scope
- **Collection:** Gather publicly available data systematically
- **Processing:** Organize and normalize collected information
- **Analysis:** Correlate data, identify patterns, and form hypotheses
- **Verification:** Cross-check sources to confirm reliability
- **Reporting:** Document findings in a structured, reproducible format



Strengths and Limitations of OSINT

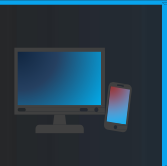
■ Strengths of OSINT

- Broad accessibility
- Rapid intelligence generation
- Cost efficiency
- Attribution support
- Evidence discovery

■ Limitations of OSINT

- Information overload
- Misinformation
- Identity spoofing
- Attribution uncertainty
- Legal boundaries





OSINT Industries

- **Law Enforcement** & Cybercrime Investigation
- **National Security** & Intelligence
- **Corporate Security** & Risk Management
- **Financial & Fraud** Investigation
- **Cybersecurity & Threat** Intelligence
- **Journalism & Investigative** Reporting
- **Legal & Compliance**
- **Humanitarian & Crisis** Response
- **Academic & Research** Communities
- **Private** Investigations



Pre-Investigations





Concept and Purpose of Pre-investigation

- **Pre-investigation** is the structured **preparation phase** conducted before any **OSINT** or cyber investigative **activity begins**
- The **purpose** is to establish a **secure operational environment**, protect **investigator identity**, and ensure **evidentiary integrity**
- The primary **objectives** include
 - Protecting **investigator identity** (Operational Security – OPSEC)
 - Reducing **digital footprint** exposure
 - Preventing target **counter-actions**
 - Preparing evidence **documentation workflows**
 - Maintaining investigative **credibility**



Operational Security (OPSEC)

- **Operational Security (OPSEC)** is the discipline of **protecting investigative operations** from **exposure**
 - **Separation** of **personal** and **investigative** systems
 - Use of **dedicated devices** and **network** connections
 - Avoiding **cross-account** contamination
 - Controlling **metadata leakage**
 - Monitoring **operational traces**



<https://www.riskpal.com/what-is-operational-security/>

Hardware and Network Preparation

- Investigators should establish a **controlled technical environment**
 - Dedicated investigation **workstation**
 - Separate **internet connectivity** (**isolated network**)
 - **Avoidance of personal account** logins
 - **Secure storage** for investigative materials
- This separation **reduces traceability** and **contamination**





Hardware and Network Preparation (Cont)

- Dedicated investigation workstation or physical computer





Hardware and Network Preparation (Cont)

- Separate internet connectivity by using **isolated SIM card**
 - Travel SIM (eSIM)
 - Data-Only SIM
 - Special SIM

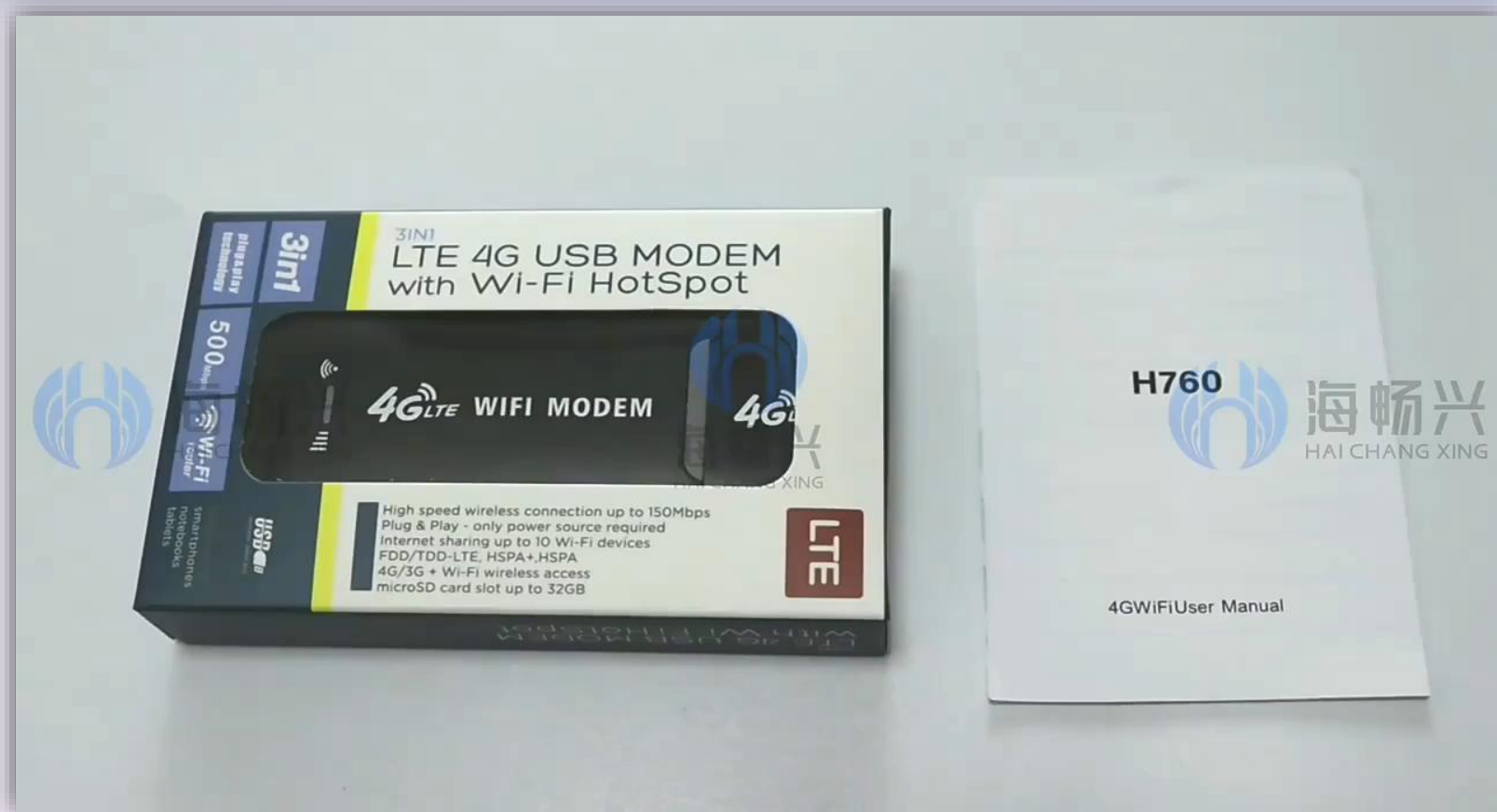


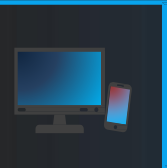


Hardware and Network Preparation (Cont)

- Isolated **device/dongle** for network connection

- Pocket WiFi
- Air Card
- USB Modem
- 4G/5G Router





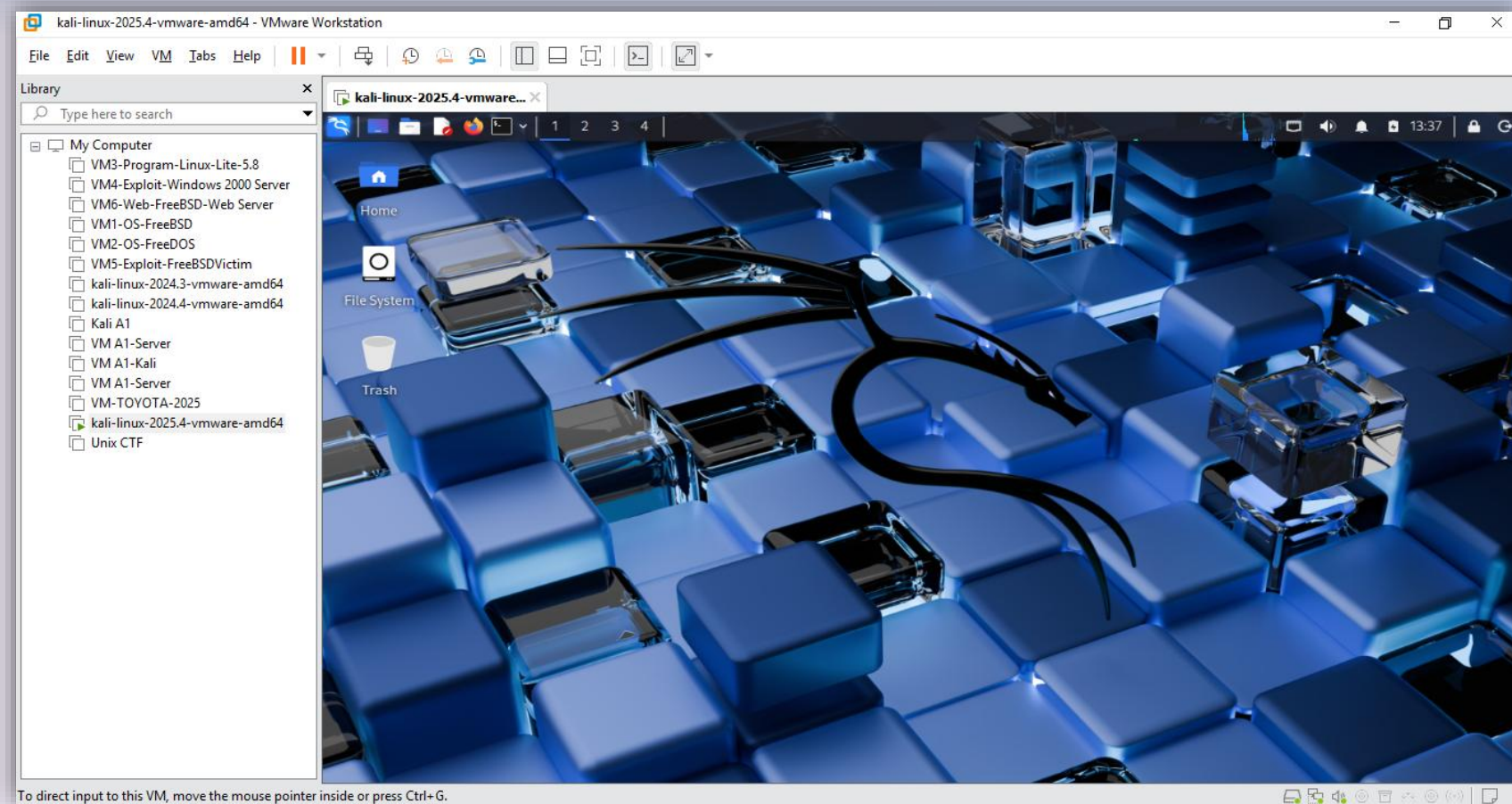
Virtual Environment Deployment

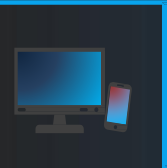
- Virtual machines (VMs) create isolated investigative environments
- Benefits for virtual environment deployment:
 - Malware containment
 - Environment rollback capability
 - Controlled experimentation
 - Evidence protection
- Recommended practices for virtual environment deployment:
 - Snapshot before investigation
 - Reset after high-risk activity
 - Separate VM profiles for different cases



Virtual Environment Deployment (Cont)

- Isolated investigative **environments** by using **Virtual Machine (VM)**
 - VMware Workstation
 - VirtualBox
 - Hyper-V





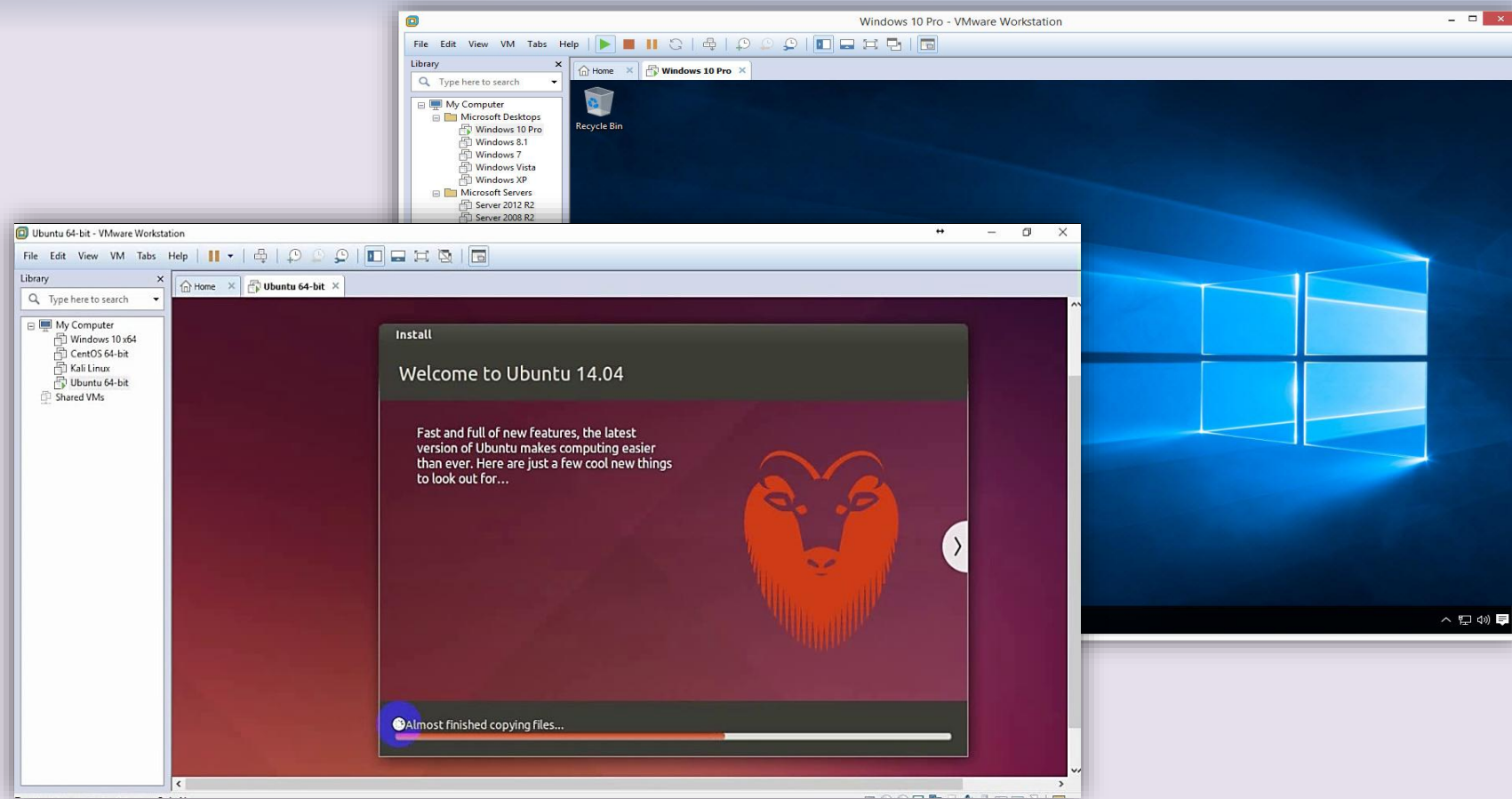
Operating System Segmentation

- Different operating environments may serve distinct purposes, no single operating environment is ideal for every task
 - General browsing and OSINT collection
 - Technical analysis and metadata review
 - Secure communication channels
- Different operating system on the host (physical machine) and the virtual machine (VM) creates layered isolation, reduces contamination risk, and improves investigative flexibility
- Segmentation reduces cross-contamination and enhances operational control



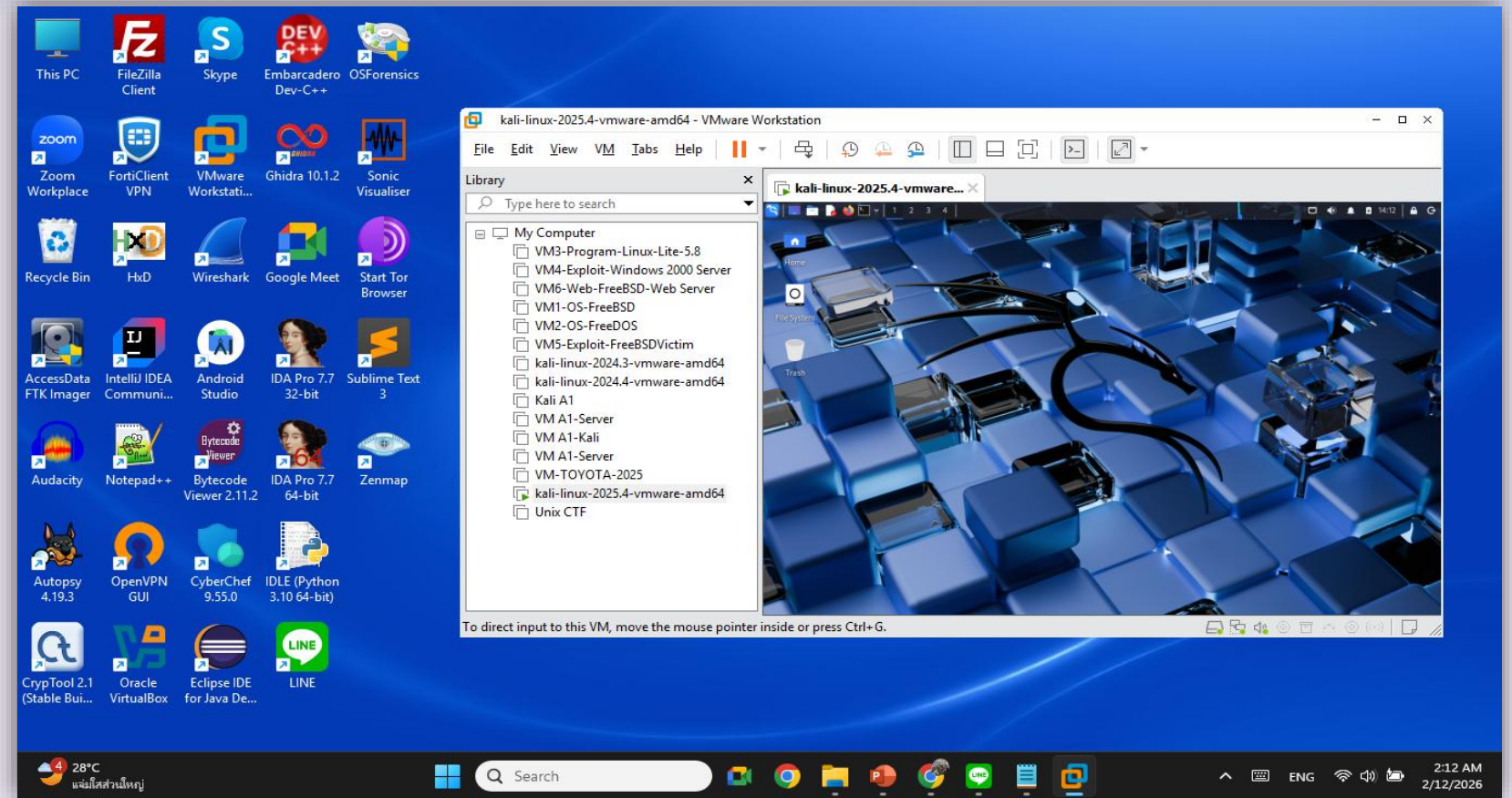
Operating System Segmentation (Cont)

- Use **one virtual machine (VM) per task/mission**
- VM-1 for Task-1
- VM-2 for Task-2
- ...
- VM-n for Task-n



Operating System Segmentation (Cont)

- Use **different operating system on the physical machine (host) and the virtual machine (VM)**
- PM → Windows
- VM → Linux





Network Anonymization Measures

- **Network anonymization** reduces exposure to monitoring
 - **VPN usage** aligned with organizational policy
 - **IP obfuscation**
 - **Controlled routing** environments
- Investigators must use **VPN** services that meet organizational standards for **data handling** and **logging** to maintain legal compliance
- By using obfuscation techniques, the target sees a **"proxy" IP** rather than the investigator's **actual address**, which protects investigator identity



Network Anonymization Measures (Cont)

- ProtonVPN
- Urban VPN
- FreeVPN
- Hide.me VPN

The screenshot displays the Proton VPN website's download page. On the left is a dark sidebar with the Proton VPN logo and a menu including Home, Subscription, Recovery, Account, Appearance, Security and privacy, and Downloads (which is highlighted). The main content area is titled 'Proton VPN clients' and includes a brief instruction: 'To secure your internet connection, download and install the Proton VPN application for your device and connect to a server.' Below this, there are seven download buttons arranged in two rows, each with a platform icon and a 'Download' button. The platforms are Android, iOS, Windows, macOS, GNU/Linux, Chromebook, and Android TV. The top right of the page features an 'Upgrade' button, the email 'wongyos@gmail.com', and a user icon 'W'. At the bottom of the sidebar, the version 'Proton VPN 5.0.299.2' is listed.

<https://protonvpn.com/>



Network Anonymization Measures (Cont)

- ProtonVPN
- **Urban VPN**
- FreeVPN
- Hide.me VPN

<https://www.urban-vpn.com/>



Network Anonymization Measures (Cont)

- ProtonVPN
- Urban VPN
- **FreeVPN**
- Hide.me VPN

The screenshot shows the FreeVPN website. At the top, a blue banner displays a notification: "Your IP 184.22.254.3 is exposed!" followed by the text "Protect your privacy with military-grade encryption." and a "Get Protection" button. Below the banner is the FreeVPN logo and a navigation menu with links for Products, Solutions, Resources, Company, and Pricing. A "Log in" link and a "Get FreeVPN" button are also present. The main content area features the heading "FreeVPN Apps" in large, bold letters. Below this, a paragraph states: "Fast, audited, WireGuard®-powered clients for all your devices. One-click protection with military-grade encryption and unlimited bandwidth." At the bottom of the main content area, there are two buttons: "Download FreeVPN" and "View Features".

<https://freevpn.com/>



Network Anonymization Measures (Cont)

- ProtonVPN
- Urban VPN
- FreeVPN
- **Hide.me VPN**

The screenshot shows the Hide.me VPN website. The header is dark blue with the Hide.me logo (a blue triangle with a white figure) and the text "HIDE me since 2012". Navigation links include "Why hide.me?", "VPN Apps", "Servers", "Pricing", "Help", "Blog", and "My Account". A yellow button says "GET HIDE.ME" and a language dropdown shows "EN". The main content area has a pink and orange background with a large heart graphic. The heart contains the text "From \$2.49 /month". Above the heart, the text "Own Your Privacy" is displayed, with "Privacy" underlined. Below it, the text "Your data deserves the best treatment" is shown. A yellow button says "Get hide.me VPN". Below this button, the text "Pricing Guarantee: Renews at the same price and duration" is visible. At the bottom, there is a section titled "Get started with hide.me VPN for FREE" with icons for Windows, macOS, Android, and iOS. To the right of this section, there are two chat buttons: "Need help? Chat with us now!" and "Chat with us now!". The URL "https://hide.me/en/" is displayed at the bottom of the screenshot.

Own Your Privacy
Your data deserves the best treatment

Get hide.me VPN

Pricing Guarantee: Renews at the same price and duration

From **\$2.49** /month

Get started with hide.me VPN for FREE

<https://hide.me/en/>

Need help? Chat with us now!

Chat with us now!



Persona Development

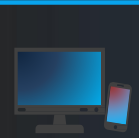
- A **persona** is a controlled **investigative identity** used to interact with online environments **without revealing** the **investigator's real identity**
 - Internal consistency (**name, background, behavior**)
 - **Realistic** digital footprint
 - **Separation from personal identity**
 - Ethical and legal compliance
- **Persona types:**
 - **Passive persona** — observation only
 - **Active persona** — controlled interaction
 - **Deep persona** — long-term infiltration



Persona Development (Cont)

- **Persona Profile Template:** a structured framework used to define and manage a controlled **investigative identity**, ensuring it **remains realistic**

Category	Description (Template)	Example Data
1. Core Identity	Name, Age, DOB, Nationality, and Gender	Name: "Alex Turner", Age: 29, Nationality: British
2. Background	Education, career history, and current residence	Former Junior Developer, currently Freelance Graphic Designer living in Berlin
3. Personality & Interests	Behavioral traits, hobbies, and specific niche interests	Tech enthusiast, interested in privacy tools and decentralized finance (DeFi)



Persona Development (Cont)

Category	Description (Template)	Example Data
4. Digital Ecosystem	Dedicated emails, social media handles, and forum memberships.	Email: a.turner.design@proton.me, X/Twitter: @ATurnerDev, Forum: Active on "Berlin-Tech-Hub".
5. Operational Setup	Browser profile, VPN region, and VM configuration.	Browser: Firefox Profile #4, VPN: Germany (Dedicated IP), Environment: Linux VM.
6. Target/Task Alignment	How this persona fits the specific investigation.	Designed to monitor leaked data forums without triggering security alerts (Passive Persona).



Persona Development (Cont)

■ Example of Persona Profile

- Full Name
- Nickname
- Sex
- DOB
- Address

Persona Template

แบบฟอร์มตัวตนปลอม

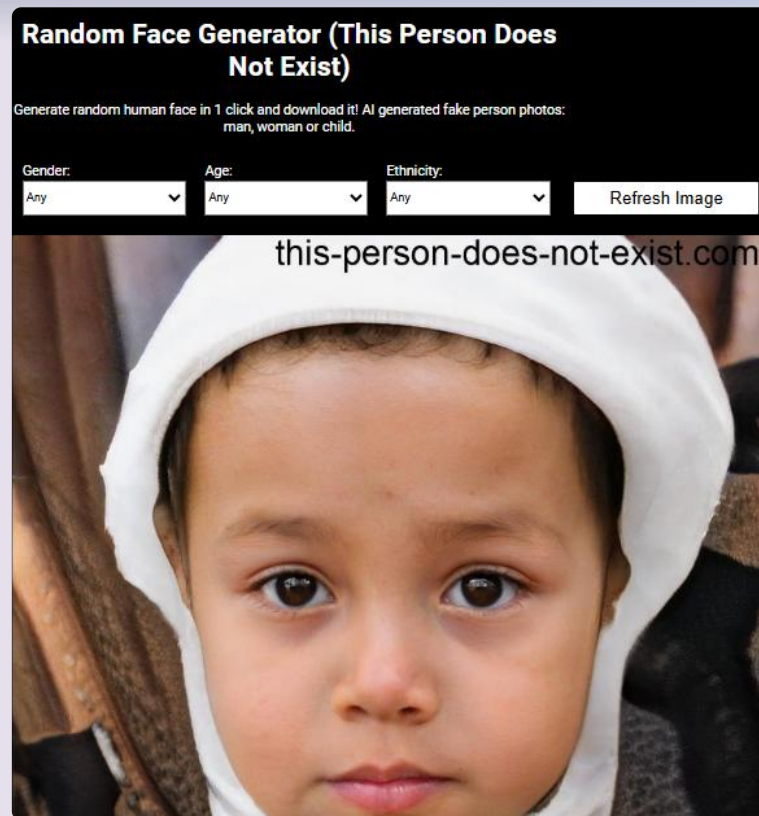
Name ชื่อ	Kongkapun Rakarndee
Sex เพศ	Male
Marital status สถานภาพ	Single
DOB วันเกิด	May 1, 1999
Place of Birth สถานที่เกิด	Bangkok, Thailand

Where do you live? คุณอาศัยอยู่ที่ไหน	90/345 Ratchadaphisek Road, Huai Khwang, Bangkok, Thailand, 10310
How long have you lived there? คุณอาศัยอยู่ที่นั่นนานเท่าไร	20 Years
Why do you live there? ทำไมคุณจึงอาศัยอยู่ที่นั่น	It is family hometown
Where did you grow up? คุณโตที่ไหน	Huai Khwang, Bangkok, Thailand

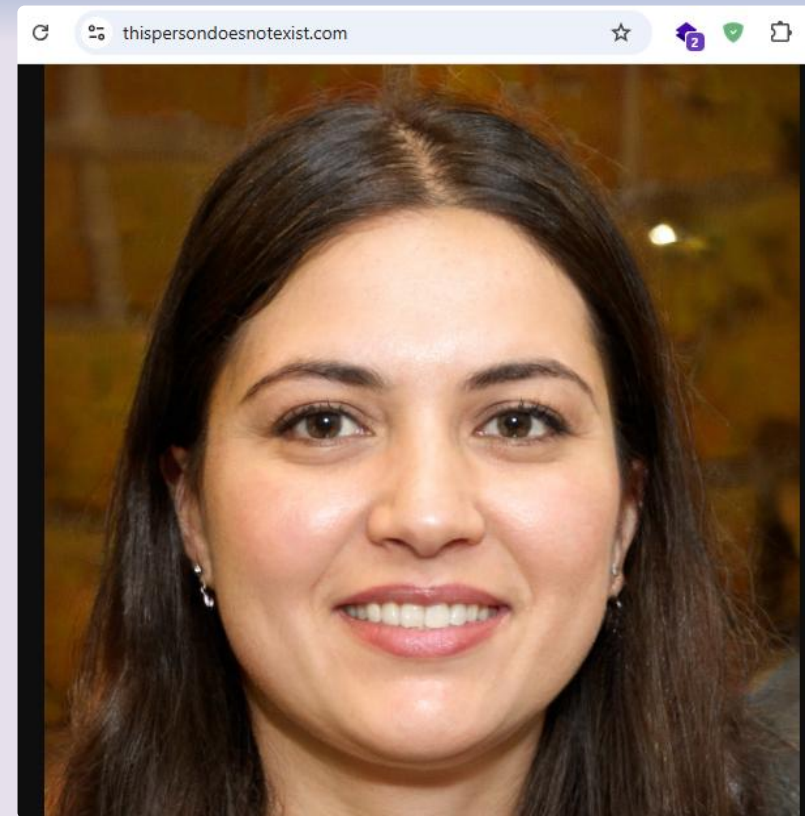


Persona Development (Cont)

- Example of **Persona Images**
 - Face
 - Full Body
- **This-Person-Does-Not-Exist.com**
- **ThisPersonDoesNotExist.com**



<https://this-person-does-not-exist.com/en>



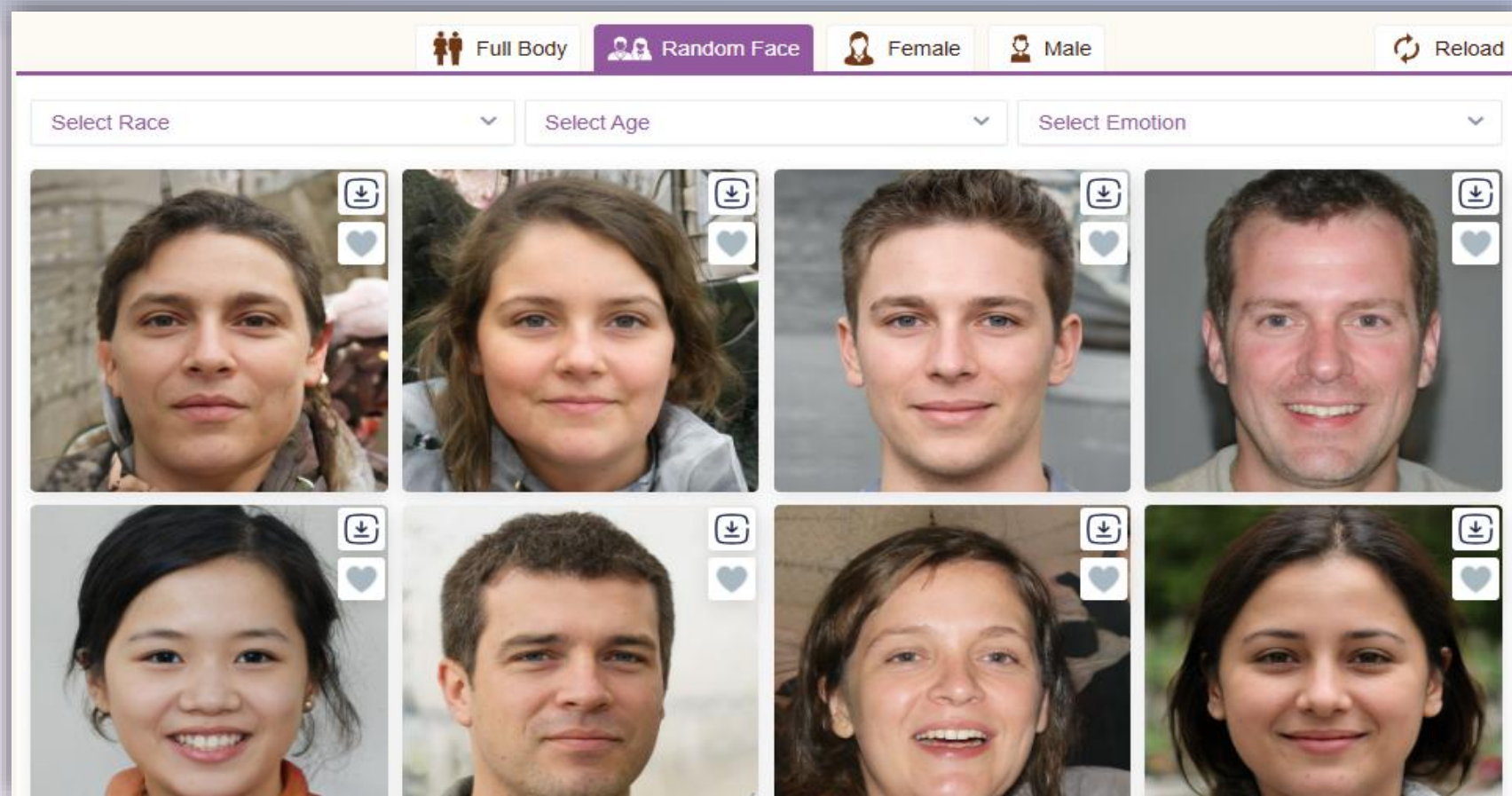
<https://www.thispersondoesnotexist.com/>



Persona Development (Cont)

- ThisPersonNot Exist.org

<https://thispersonnotexist.org/>





Avatar Ecosystem Creation

- **Avatar ecosystem creation** is the process of building a credible **digital presence** to **support a persona**, ensuring it appears authentic to targets and provides the necessary access for the investigation
- A credible persona requires a **supporting digital presence**
 - **Dedicated Emails**: Foundation for registrations; strictly isolated from personal/work accounts
 - **Social Media Profiles**: Target-relevant accounts (e.g., LinkedIn, X) to build visible history
 - **Behavioral History**: Timed logs of likes, posts, and shares to simulate human activity
 - **Network of Connections**: Followers and friends to avoid the suspicion of a "**zero-friend**" profile

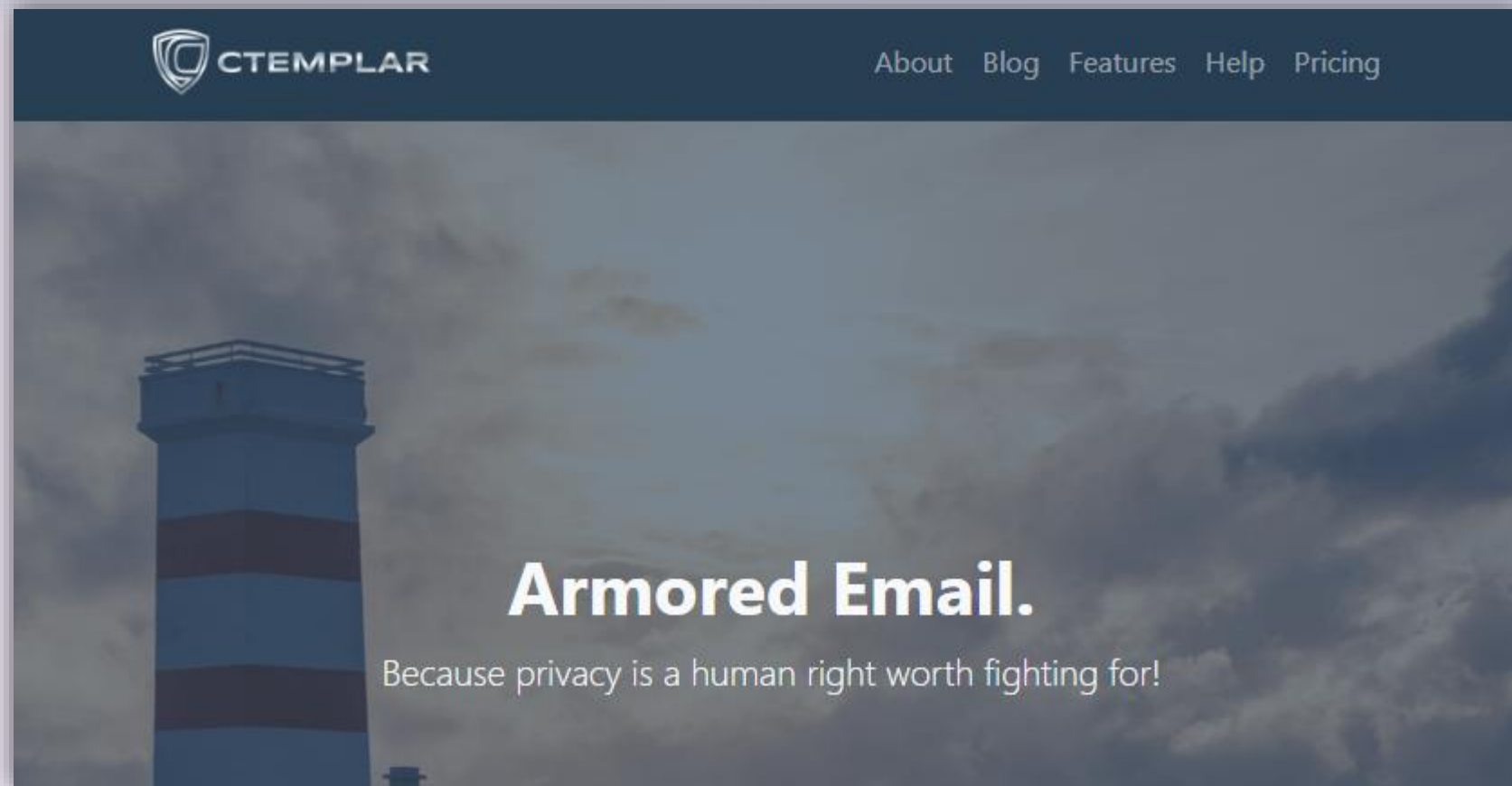


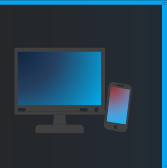
Avatar Ecosystem Creation (Cont)

- Armored **Email Avatar**

- CTemplar mail
- Proton Mail

<https://ctemplar.com/>



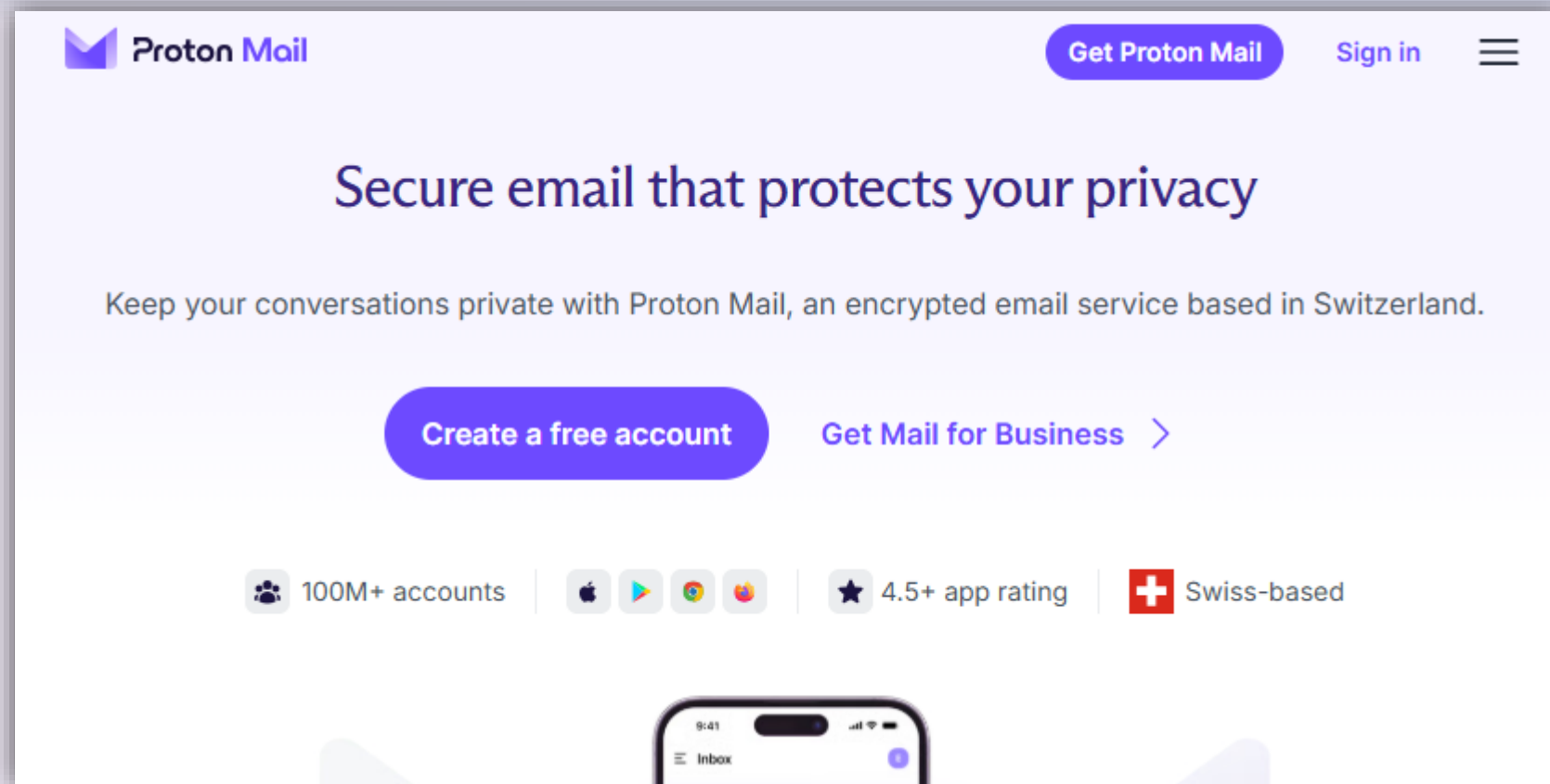


Avatar Ecosystem Creation (Cont)

- Armored **Email Avatar**

- CTemplar mail
- **Proton Mail**

<https://proton.me/mail>





Avatar Ecosystem Creation (Cont)

- Social Media Avatar

The screenshot shows a Facebook profile for Kongkapan Panyadee. The profile picture is a circular portrait of a man with dark hair and a slight smile. To the right of the picture, the name 'Kongkapan Panyadee' is displayed in bold. Below the name, it says '868 friends • 130 mutual' and 'ผมชื่อป้านครับ' (My name is Puan). Further down, location and education details are listed: 'Bangkok', 'Self-Employed', and 'Thammasat University'. There are buttons for 'Add friend' and 'Message'. Below the profile header, there are tabs for 'All', 'About', 'Friends', 'Photos', 'Reels', and 'More'. The 'All' tab is selected. The main content area is divided into two sections: 'Personal details' on the left and 'Posts' on the right. The 'Personal details' section lists 'Lives in Bangkok, Thailand', 'From Bangkok, Thailand', and 'November 25, 1983'. The 'Posts' section shows a post from Kongkapan Panyadee dated September 24, 2023, with the text 'อยากจะสนุกก็ต้องมาทำความรู้จักกัน อยากซื้อดอกไม้ส่งมาได้เลย ข่ายถูก ๆ'.

Kongkapan Panyadee
868 friends • 130 mutual
ผมชื่อป้านครับ
📍 Bangkok 🏢 Self-Employed 🎓 Thammasat University

[Add friend](#) [Message](#)

[All](#) [About](#) [Friends](#) [Photos](#) [Reels](#) [More](#)

Personal details

- 📍 Lives in Bangkok, Thailand
- 🏠 From Bangkok, Thailand
- 🎂 November 25, 1983

Posts [Filters](#)

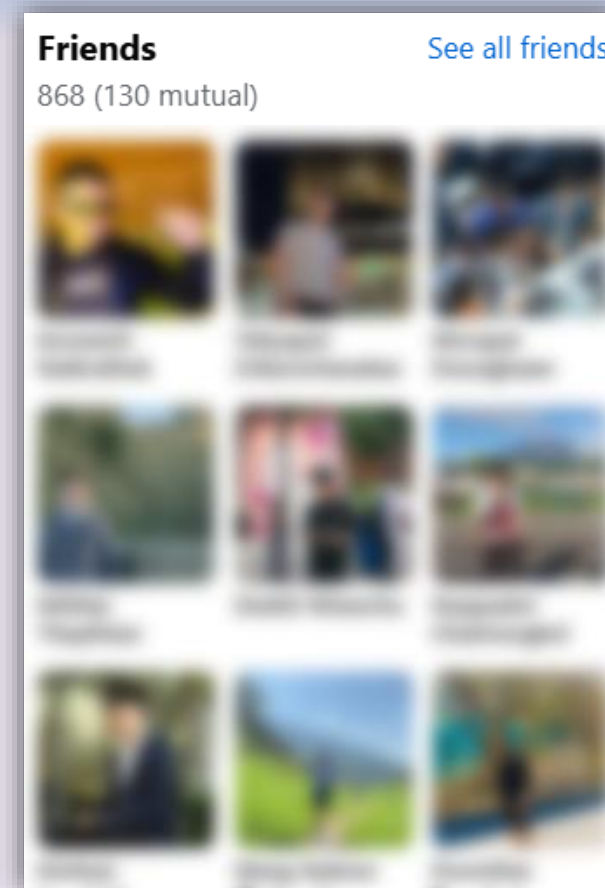
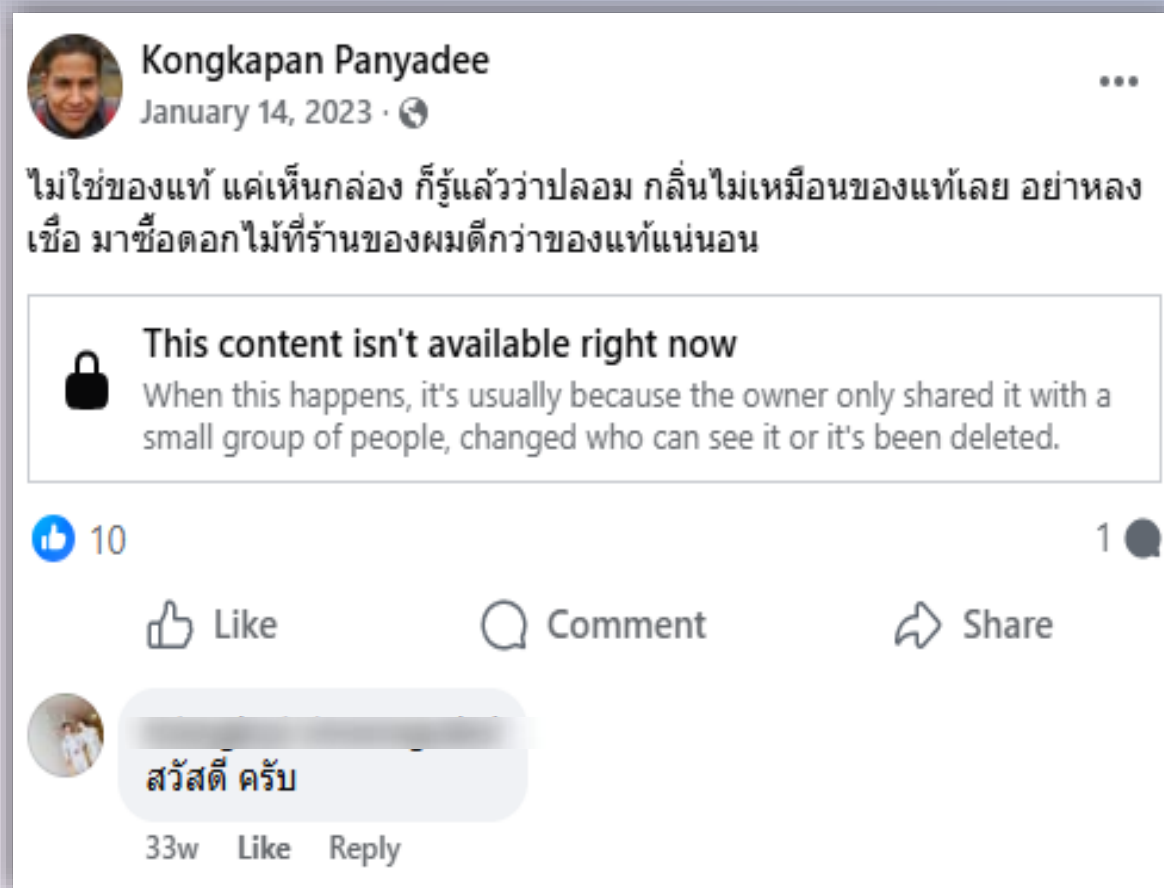
Kongkapan Panyadee
September 24, 2023 · 🌐

อยากจะสนุกก็ต้องมาทำความรู้จักกัน อยากซื้อดอกไม้ส่งมาได้เลย ข่ายถูก ๆ



Avatar Ecosystem Creation (Cont)

- **Behavioral History** for Avatar





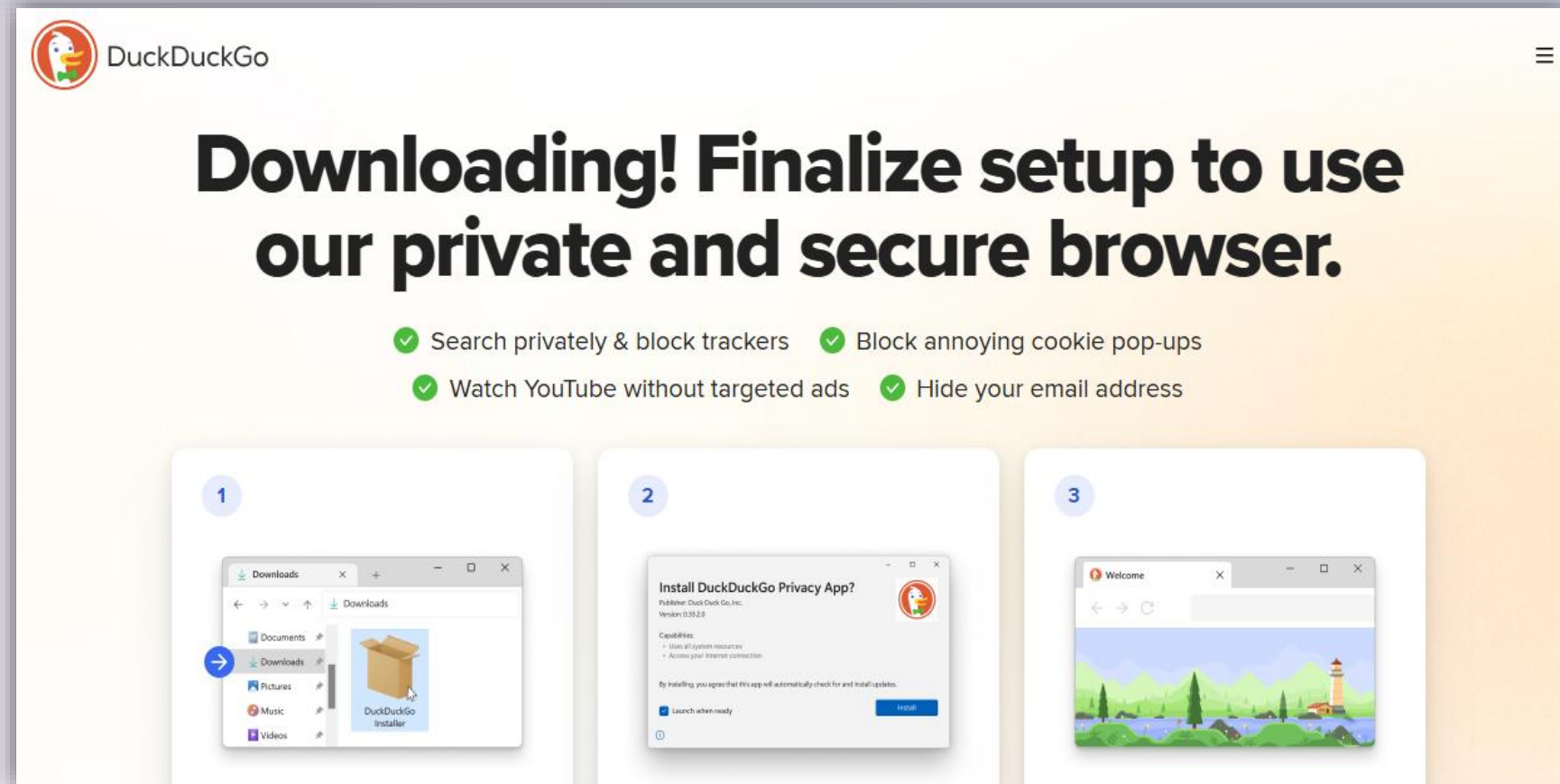
Digital Footprint Management

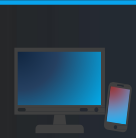
- **Digital Footprint Management** is the discipline of **controlling** and **minimizing** the **operational traces left** by an investigator to prevent attribution
- The specific practices required for effective **footprint management**:
 - **Separate Browser Profiles**: Isolate cookies and history by case to prevent data leakage
 - **Unique Credentials**: Use exclusive logins for every account to block cross-tracking
 - **Metadata Awareness**: Scrub hidden file data (EXIF/GPS) to prevent identity exposure
 - **No Identity Reuse**: Never recycle persona elements across different investigations
 - **Footprint Discipline**: Avoid accidental personal logins on investigative systems



Digital Footprint Management (Cont)

- Switch to **DuckDuckGo** for private browser





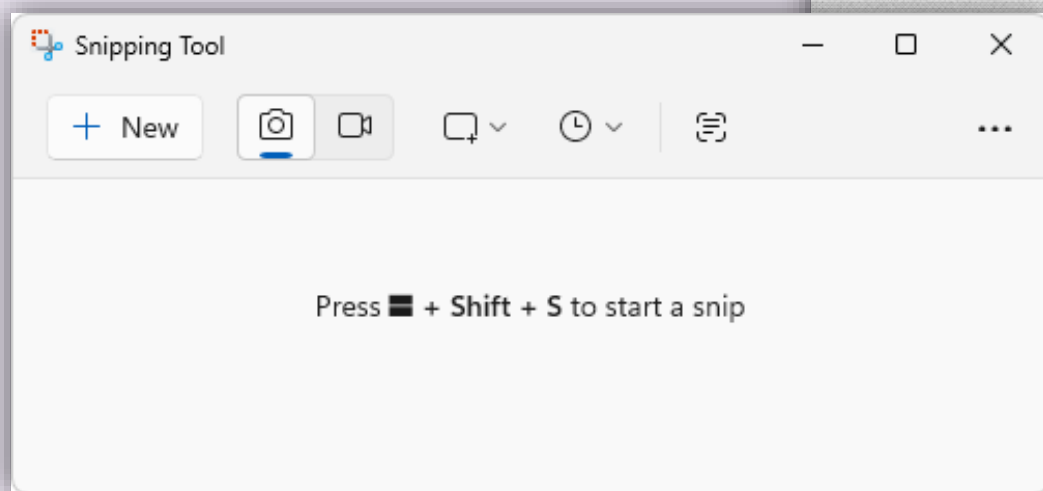
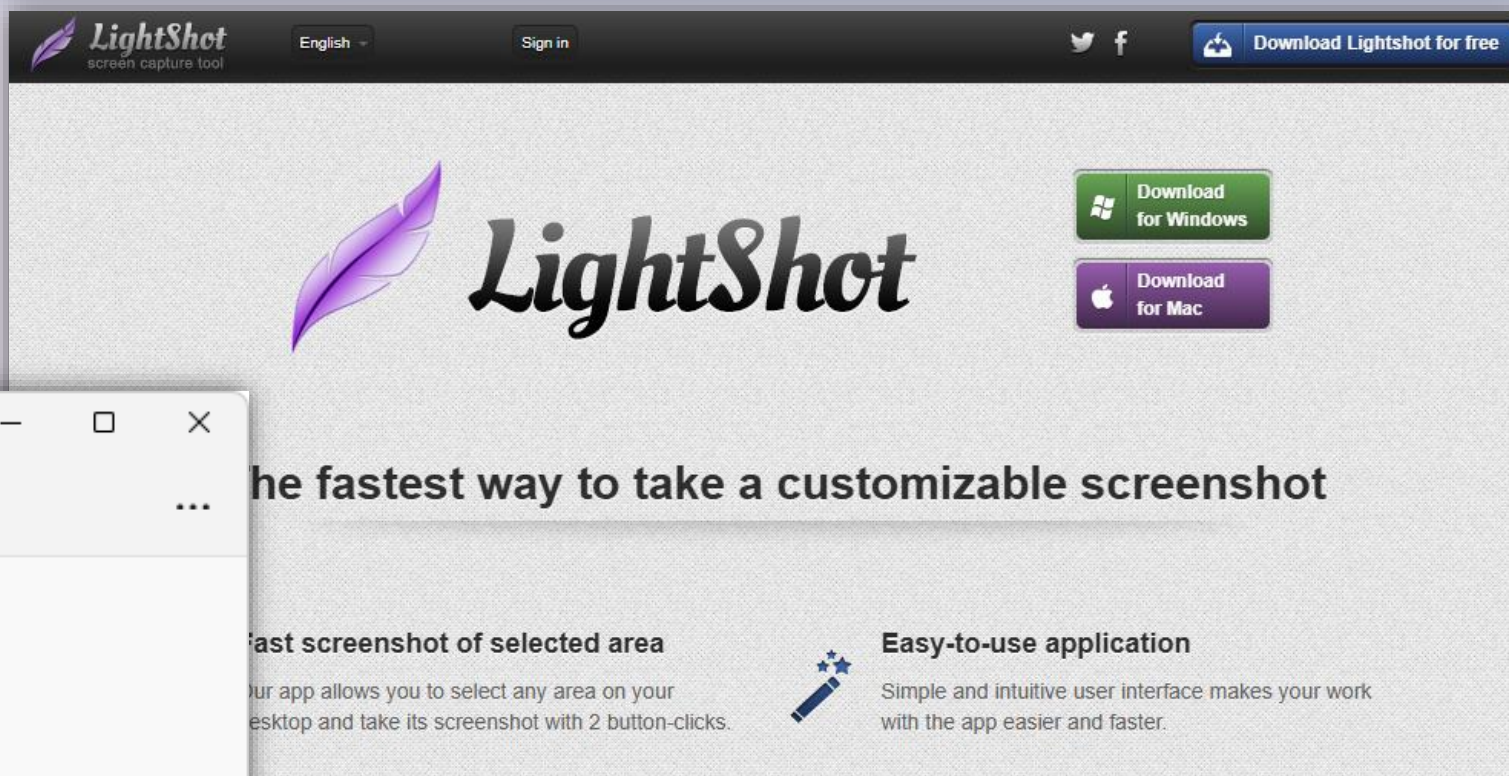
Evidence Readiness Planning

- **Before** investigation **begins**, evidence handling procedures **must be prepared**
- Essential elements:
 - Screenshot and capture tools
 - Timestamp documentation
 - URL logging
 - Case notes
 - Chain-of-custody awareness
- Preparation ensures collected intelligence remains **admissible** and **verifiable**



Evidence Readiness Planning (Cont)

- Use the **screenshot** and **capture** tools
 - Snipping Tools
 - LightShot

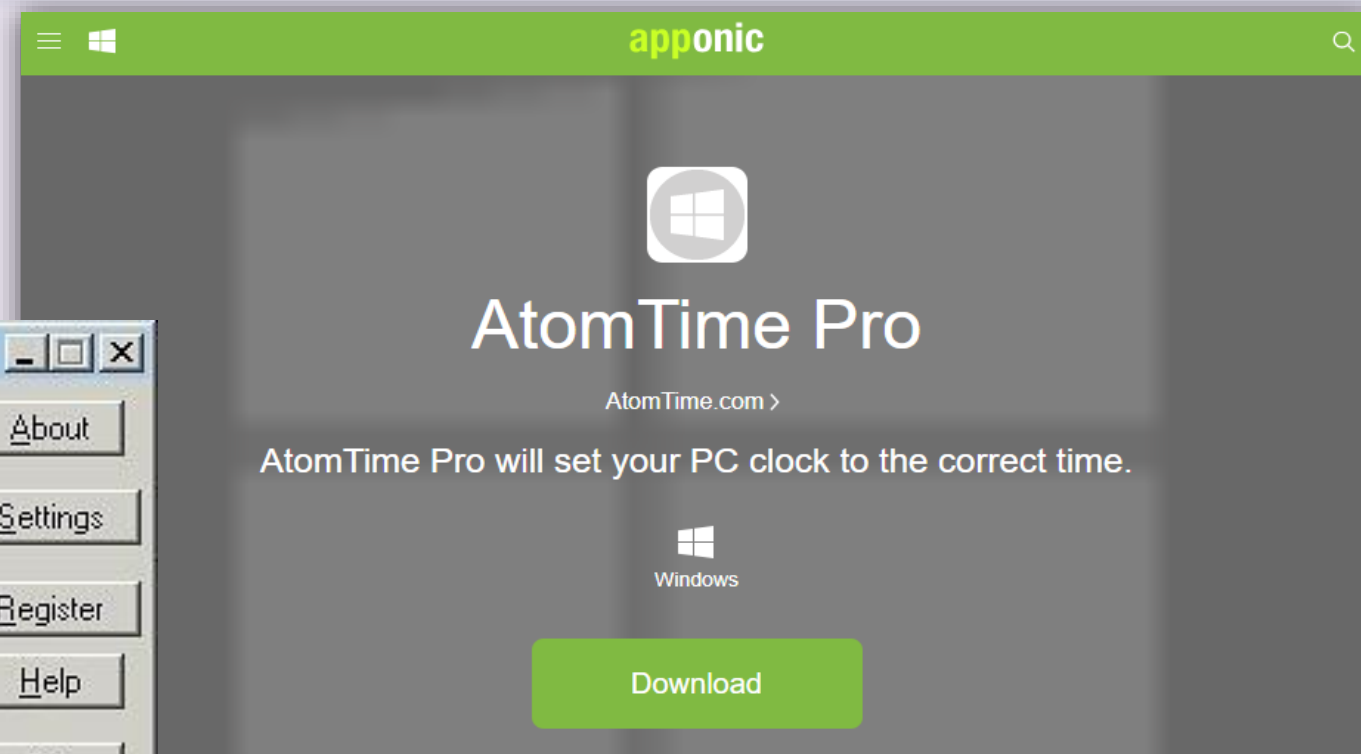


<https://app.prntscr.com/en/index.html>



Evidence Readiness Planning (Cont)

- **Timestamp**: show the **correct time** and **universal time** on PC clock



<https://atomtime-pro.apponic.com/>



Risk Assessment

- **Risk assessment** is the structured process of identifying, analyzing, and mitigating **operational risks** in the **initiating investigative actions**
- The **purpose** of risk assessment is **not to eliminate all risk** but to **understand risk** and make informed **operational decisions**
- Evaluate potential operational risks:
 - **Target retaliation capability**
 - **Exposure likelihood**
 - **Malware threats**
 - **Legal implications**

Q & A

Asst.Prof.Pol.Lt.Col.Dr. Wongyos Keardsri (Bank)

Lecturer (Level 3) | Faculty of Forensic Science | Royal Police Cadet Academy | Thailand

Email: wongyos@gmail.com | Line ID: wongyos | Phone/WhatsApp: +6689-599-3490

Facebook: facebook.com/wongyos



End